# Visvesvaraya Technological University

**BELAGAVI, KARNATAKA**

ವಿಶ್ವೇಶ್ವರಯ್ಯ ತಾಂತ್ರಿಕ ವಿಶ್ವವಿದ್ಯಾಲಯ

ಬೆಳಗಾವಿ, ಕರ್ನಾಟಕ

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ASSIGNMENT REPORT

# “MACHINE LEARNING (BRI405B)”

PROBLEM STATEMENT:
“DIABETES HEALTHCARE INDICATOR”


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**2025-26.**

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## 1.ABOUT DATASET:

Dataset source:https://archive-beta.ics.uci.edu/dataset/891/cdc+diabetes+health+indicators

The CDC Diabetes Health Indicators dataset is used for predicting diabetes based on various health and lifestyle conditions. The dataset was collected from the Behavioral Risk Factor Surveillance System (BRFSS) conducted by the Centers for Disease Control and Prevention (CDC).

Diabetes is one of the most common chronic diseases worldwide. Early identification of diabetes risk factors can help healthcare professionals take preventive measures and improve patient outcomes. Therefore, this dataset is highly significant for developing machine learning models that can predict diabetes based on health indicators.

The dataset contains information related to:

- Blood pressure
- Cholesterol
- BMI
- Smoking habits
- Physical activity
- Age
- General health
- Mental health
- Physical health

Machine learning techniques are used to analyze these factors and predict whether a person is diabetic or non-diabetic.

## 2.DATA INFORMATION:

The dataset contains 253680 records and 22 features. The target variable is Diabetes_binary. Dataset Information:

- Number of Rows: 253680
- Number of Columns: 22

- Dataset Type: Healthcare Dataset
- Machine Learning Type: Supervised Learning
- Missing Values: No
- Target Variable: Diabetes_binary

## 3.VARIABLE INFORMATION OF DATASET:

The dataset contains multiple health-related variables.

- HighBP – High blood pressure indicator.
- HighChol – High cholesterol indicator.
- CholCheck – Cholesterol check history.
- BMI – Body Mass Index value.
- Smoker – Smoking habit indicator.
- Stroke – Stroke history.
- PhysActivity – Physical activity condition.
- Fruits – Fruit consumption.
- Veggies – Vegetable consumption.
- GenHlth – General health status.
- MentHlth – Number of days Mental health was not good • PhysHlth –Number of days Physical Health was not good.
- Age – Age category.
- Income – Income category.

## 4.TOOLS USED:

The following tools were used:

- Python Programming Language
- Google Collab

- Pandas Library
- NumPy Library
- Matplotlib Library
- Seaborn Library
- Scikit-learn Library Google Collab was used for implementation because it provides a cloudbased Python environment that is suitable for machine learning experiments.

DATA PREPROCESSING:

Data preprocessing is one of the most important steps in machine learning. Raw datasets often contain Inconsistencies, missing values, and irrelevant information. The following preprocessing steps were performed:

- Loading the dataset using Pandas.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

5.EXPLORATORY DATA ANALYSIS:

Exploratory Data Analysis was performed to understand the structure and characteristics of the dataset.

Dataset Inspection:

The dataset was loaded using Pandas and inspected using:

- df.head()

```python
df.head()
```

| | Diabetes_binary | HighBP | HighChol | CholCheck | BMI | Smoker | Stroke | HeartDiseaseorAttack | PhysActivity | Fruits | ... |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0.0 | 1.0 | 1.0 | 1.0 | 40.0 | 1.0 | 0.0 | 0.0 | 0.0 | 0.0 | ... |
| 1 | 0.0 | 0.0 | 0.0 | 0.0 | 25.0 | 1.0 | 0.0 | 0.0 | 1.0 | 0.0 | ... |
| 2 | 0.0 | 1.0 | 1.0 | 1.0 | 28.0 | 0.0 | 0.0 | 0.0 | 0.0 | 1.0 | ... |
| 3 | 0.0 | 1.0 | 0.0 | 1.0 | 27.0 | 0.0 | 0.0 | 0.0 | 1.0 | 1.0 | ... |
| 4 | 0.0 | 1.0 | 1.0 | 1.0 | 24.0 | 0.0 | 0.0 | 0.0 | 1.0 | 1.0 | ... |

5 rows × 22 columns

| AnyHealthcare | NoDocbcCost | GenHlth | MentHlth | PhysHlth | DiffWalk | Sex | Age | Education | Income |
|---|---|---|---|---|---|---|---|---|---|
| 1.0 | 0.0 | 5.0 | 18.0 | 15.0 | 1.0 | 0.0 | 9.0 | 4.0 | 3.0 |
| 0.0 | 1.0 | 3.0 | 0.0 | 0.0 | 0.0 | 0.0 | 7.0 | 6.0 | 1.0 |
| 1.0 | 1.0 | 5.0 | 30.0 | 30.0 | 1.0 | 0.0 | 9.0 | 4.0 | 8.0 |
| 1.0 | 0.0 | 2.0 | 0.0 | 0.0 | 0.0 | 0.0 | 11.0 | 3.0 | 6.0 |
| 1.0 | 0.0 | 2.0 | 3.0 | 0.0 | 0.0 | 0.0 | 11.0 | 5.0 | 4.0 |

- Checking data information using df.info().

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 253680 entries, 0 to 253679
Data columns (total 22 columns):
 #   Column                Non-Null Count   Dtype  
---  ------                --------------   -----  
 0   Diabetes_binary       253680 non-null  float64
 1   HighBP                253680 non-null  float64
 2   HighChol              253680 non-null  float64
 3   CholCheck             253680 non-null  float64
 4   BMI                   253680 non-null  float64
 5   Smoker                253680 non-null  float64
 6   Stroke                253680 non-null  float64
 7   HeartDiseaseorAttack  253680 non-null  float64
 8   PhysActivity          253680 non-null  float64
 9   Fruits                253680 non-null  float64
 10  Veggies               253680 non-null  float64
 11  HvyAlcoholConsump     253680 non-null  float64
 12  AnyHealthcare         253680 non-null  float64
 13  NoDocbcCost           253680 non-null  float64
 14  GenHlth               253680 non-null  float64
 15  MentHlth              253680 non-null  float64
 16  PhysHlth              253680 non-null  float64
 17  DiffWalk              253680 non-null  float64
 18  Sex                   253680 non-null  float64
 19  Age                   253680 non-null  float64
 20  Education             253680 non-null  float64
 21  Income                253680 non-null  float64
dtypes: float64(22)
memory usage: 42.6 MB
```

- Statistical analysis using df.describe().

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 253680 entries, 0 to 253679
Data columns (total 22 columns):
 #   Column                Non-Null Count   Dtype  
---  ------                --------------   -----  
 0   Diabetes_binary       253680 non-null  float64
 1   HighBP                253680 non-null  float64
 2   HighChol              253680 non-null  float64
 3   CholCheck             253680 non-null  float64
 4   BMI                   253680 non-null  float64
 5   Smoker                253680 non-null  float64
 6   Stroke                253680 non-null  float64
 7   HeartDiseaseorAttack  253680 non-null  float64
 8   PhysActivity          253680 non-null  float64
 9   Fruits                253680 non-null  float64
 10  Veggies               253680 non-null  float64
 11  HvyAlcoholConsump     253680 non-null  float64
 12  AnyHealthcare         253680 non-null  float64
 13  NoDocbcCost           253680 non-null  float64
 14  GenHlth               253680 non-null  float64
 15  MentHlth              253680 non-null  float64
 16  PhysHlth              253680 non-null  float64
 17  DiffWalk              253680 non-null  float64
 18  Sex                   253680 non-null  float64
 19  Age                   253680 non-null  float64
 20  Education             253680 non-null  float64
 21  Income                253680 non-null  float64
dtypes: float64(22)
memory usage: 42.6 MB
```

- Verifying missing values.

```
df.isnull().sum()
```

| | 0 |
|---|---|
| Diabetes_binary | 0 |
| HighBP | 0 |
| HighChol | 0 |
| CholCheck | 0 |
| BMI | 0 |
| Smoker | 0 |
| Stroke | 0 |
| HeartDiseaseorAttack | 0 |
| PhysActivity | 0 |
| Fruits | 0 |
| Veggies | 0 |
| HvyAlcoholConsump | 0 |
| AnyHealthcare | 0 |
| NoDocbcCost | 0 |
| GenHlth | 0 |
| MentHlth | 0 |
| PhysHlth | 0 |
| DiffWalk | 0 |
| Sex | 0 |
| Age | 0 |
| Education | 0 |
| Income | 0 |

- Feature and target separation.
- Train-test splitting. The dataset did not contain missing values, so additional cleaning was not Required.

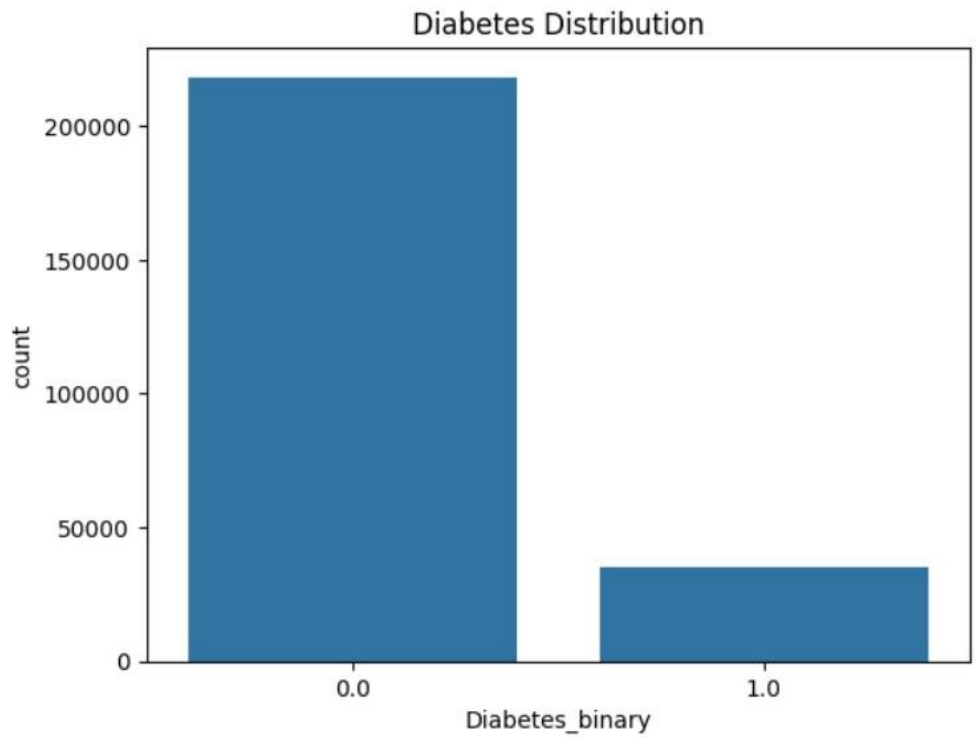
**Diabetes distribution :**

```
sns.countplot(x='Diabetes_binary', data=df)
plt.title("Diabetes Distribution")
plt.show()
```

Majority of records belong to non-diabetic classes.

Dataset is slightly imbalanced.

..




Fig,:diabetes distribution plot

Observation:

The plot shows the distribution of diabetic and non-diabetic individuals in the dataset. The number of non-diabetic records is higher compared to diabetic records, indicating that the dataset is slightly imbalanced.

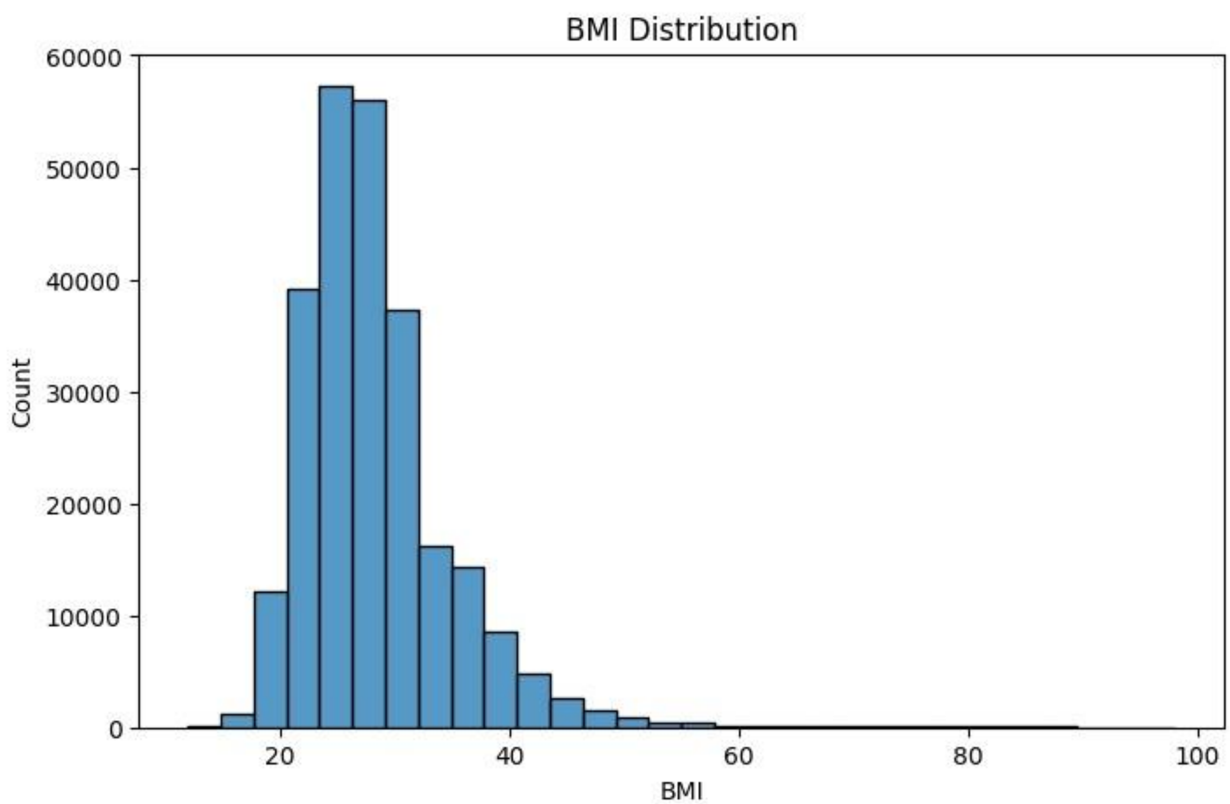
**BMI DISTRIBUTION:**

```
plt.figure(figsize=(8,5))
sns.histplot(df['BMI'], bins=30)
plt.title("BMI Distribution")
plt.show()
```

Most BMI values are concentrated in the normal and overweight categories. Higher BMI values are associated with increased risk of diabetes.

Normal BMI=18.5-24.9

Overweight=25-29



## CORRELATION HEATMAP:

A heatmap is a graphical representation of data in which values are represented using different colors.

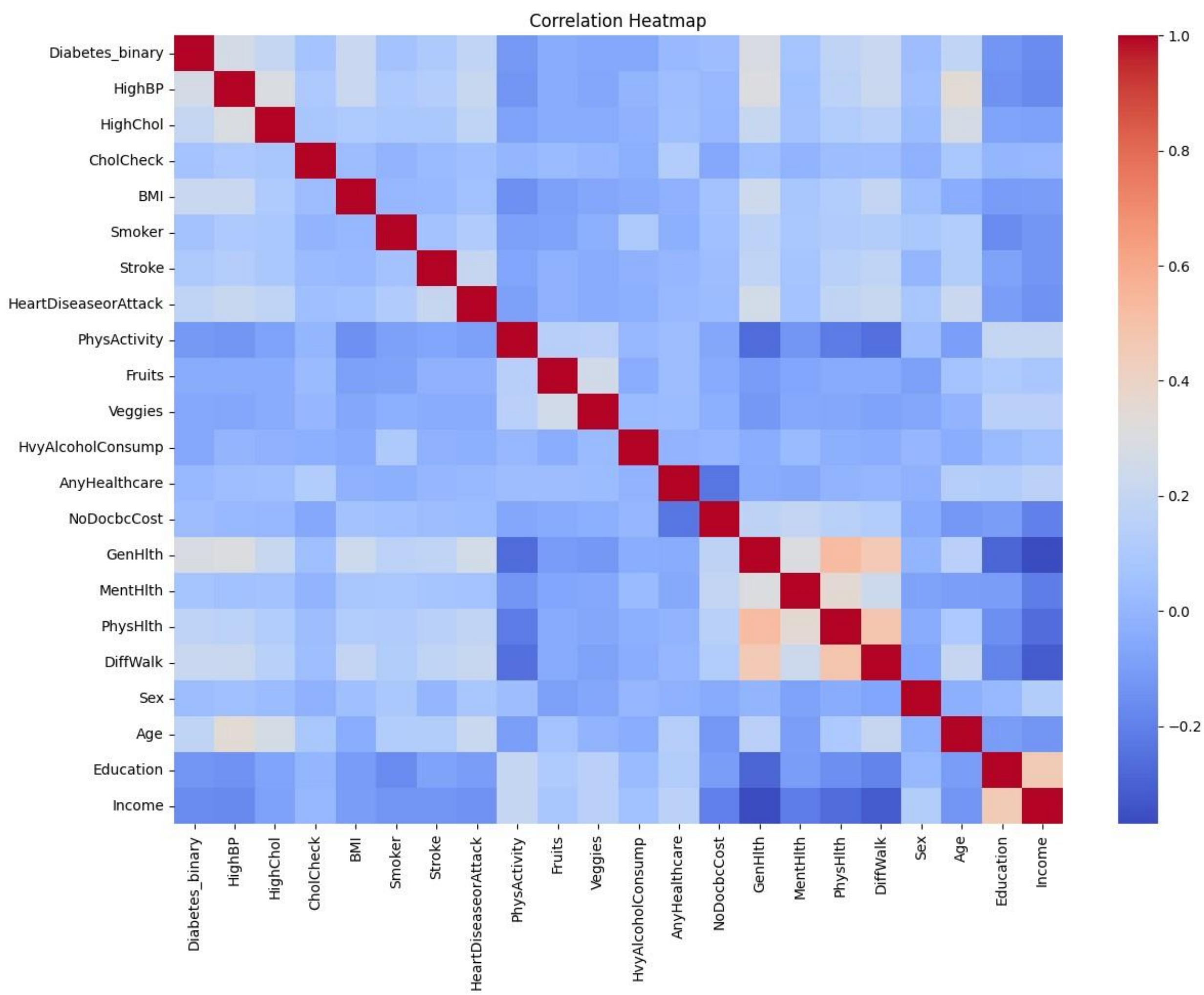
It is used to visualize the relationship between different features in a dataset.

```
plt.figure(figsize=(14,10))
sns.heatmap(df.corr(), cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```

Cool colors(blue)-negative correlation

Warm color(red)-positive correlation

White/light colors -correlation close to 0.



The red squares running diagonally from top-left to bottom-right represent a variable correlated with itself.

Example:

BMI vs BMI = 1

Age vs Age = 1

Income vs Income = 1

GenHlth, MentHlth, PhysHlth, and DiffWalk show moderate positive correlation with each other.

People with poor general health often report poor mental and physical health.

Most other features show light colors.This indicates a weak correlation between many variables.

Significance of the Correlation Heatmap:

The correlation heatmap was used to identify relationships between health indicators and diabetes risk. Rather than serving only as a visualization tool, the heatmap helps determine which variables are most strongly associated with diabetes and whether these relationships are consistent with known medical knowledge.

BMI (Body Mass Index):

BMI showed one of the strongest positive relationships with diabetes. This observation is medically meaningful because obesity is a major risk factor for Type 2 Diabetes. Excess body fat can reduce insulin sensitivity, making it more difficult for the body to regulate blood glucose levels. Therefore, individuals with higher BMI values are more likely to develop diabetes.

Age:

Age demonstrated a positive relationship with diabetes. As people grow older, insulin sensitivity often decreases and the likelihood of developing chronic diseases increases. Therefore, the positive association between age and diabetes observed in the heatmap is consistent with medical understanding.

HighBP (High Blood Pressure):

High blood pressure was positively related to diabetes. Hypertension and diabetes frequently occur together because both are associated with metabolic disorders and unhealthy lifestyle factors. The heatmap supports this well-established relationship.

HighChol (High Cholesterol):

High cholesterol showed a positive relationship with diabetes. Individuals with diabetes often experience abnormal lipid profiles and increased cardiovascular risk. Therefore,the association observed in the heatmap is medically reasonable.

GenHlth (General Health):

General health status was positively associated with diabetes risk. Poor general health may indicate the presence of chronic medical conditions, unhealthy lifestyle habits, or reduced physical well-being, all of which contribute to diabetes development.

DiffWalk (Difficulty Walking):

Difficulty walking also showed a positive relationship with diabetes. Reduced mobility may result from obesity, aging, cardiovascular problems, or diabetic complications. Therefore, individuals reporting difficulty walking are more likely to be affected by diabetes.

Overall Interpretation:

The heatmap indicates that BMI, Age, HighBP, HighChol, GenHlth, and DiffWalk are among the most relevant features associated with diabetes risk. These findings are consistent with established medical knowledge regarding diabetes risk factors. Therefore, the heatmap not only visualizes correlations but also validates that the dataset contains meaningful healthcare indicators for diabetes prediction.

## 6.MODEL IMPLEMENTATION:

- **LOGISTIC REGRESSION**
- **DECISION TREE**
- **RANDOM FOREST**

## LOGISTIC REGRESSION:

Logistic Regression is a supervised machine learning algorithm used for binary classification.

Logistic Regression was selected because:

The target variable is binary.

It is simple and it trains quickly.

It provides good baseline performance.

MODEL TRAINING:

```
from sklearn.linear_model import LogisticRegression
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

```
LogisticRegression
LogisticRegression(max_iter=1000)
```

(why 1000: a large dataset may need more iterations to find best solution)

PREDICTON:

```
y_pred = model.predict(X_test)
print(y_pred)
```

```
[0. 0. 0. ... 0. 0. 0.]
```

It predicts up to first 10 outputs.

ACCURACY:

```
from sklearn.metrics import accuracy_score
accuracy_score(y_test, y_pred)
```

```
0.865874329864396
```

Accuracy=86%.

CONFUSION MATRIX:

```
from sklearn.metrics import confusion_matrix
confusion_matrix(y_test, y_pred)
```

```
array([[42776,   963],
       [ 5842,  1155]])
```

42776(true negative)-correctly predicted nondiabetic cases.

963(false positive)-predicted diabetic but actually non-diabetic.

5842(false negative)-predicted nondiabetic but diabetic.

1155(true positive)-correctly predicted diabetic cases.

The confusion matrix reveals that the Logistic Regression model correctly classified 42,776 non-diabetic cases and 1,155 diabetic cases. However, it incorrectly classified 5,842 diabetic individuals as non-diabetic (False Negatives). This indicates that the model is biased toward the majority non-diabetic class due to class imbalance in the dataset. Although the overall accuracy is high, the large number of false negatives results in low recall for diabetic cases, which is a significant limitation in healthcare prediction systems.

CLASSIFICATION REPORT:

```
from sklearn.metrics import classification_report
print(classification_report(y_test, y_pred))
```

```
...              precision    recall  f1-score   support

         0.0       0.88      0.98      0.93     43739
         1.0       0.55      0.17      0.25      6997

    accuracy                           0.87     50736
   macro avg       0.71      0.57      0.59     50736
weighted avg       0.83      0.87      0.83     50736
```

The classification report was used to evaluate the Logistic Regression model using precision, recall, and F1-score metrics. The model achieved high performance for non-diabetic classification, while diabetic classification performance was comparatively lower due to dataset imbalance. The overall accuracy of the model was approximately 86–87%.

NON DIABETIC CLASS(0):

Precision = 0.88 means 88% of cases predicted as non-diabetic were actually non-diabetic.

Recall = 0.98 means the model correctly identified 98% of all non-diabetic individuals.

F1-score = 0.93 indicates very good performance for the non-diabetic class.

DIABETIC CLASS(1):

Precision = 0.55 means only 55% of predicted diabetic cases were actually diabetic.

Recall = 0.17 means the model identified only 17% of actual diabetic individuals.

F1-score = 0.25 indicates weak overall performance for diabetic detection.

The Logistic Regression model achieved high overall accuracy; however, the classification report revealed an important limitation. The model performed exceptionally well in identifying non-diabetic individuals but struggled to detect diabetic cases. The recall for the diabetic class was approximately 0.17, indicating that a large proportion of diabetic patients were incorrectly classified as non-diabetic. From a healthcare perspective, this is a significant concern because false negatives may delay diagnosis and treatment. Therefore, model evaluation should not depend on only accuracy, and greater emphasis should be placed on recall and F1-score for the diabetic class.

Macro Average:

Macro Average calculates the performance metric for each class separately and then computes their simple arithmetic mean. It gives equal importance to both diabetic and non-diabetic classes regardless of the number of samples present in each class.

In this study, the macro average values were:

Precision = 0.71

Recall = 0.57

F1-score = 0.59

The relatively lower macro average values indicate that the model's performance is not balanced across both classes. Although the model performs very well for non-diabetic cases, its performance for diabetic case detection is considerably lower.

Weighted Average:

Weighted Average calculates the performance metric for each class and then computes the average by considering the number of samples (support) in each class.

In this study, the weighted average values were:

Precision = 0.83

Recall = 0.87

F1-score = 0.83

The weighted average values are higher because the dataset contains a significantly larger number of non-diabetic records than diabetic records. Since the model performs well on the majority non-diabetic class, the weighted average and overall accuracy appear high

## DECISION TREE:

Decision Tree Classifier uses a tree structure to make decisions based on feature values.

Decision Tree was selected because:

Easy to understand and visualize.

Handles nonlinear relationships.

MODEL TRAINING:

```
dt_model.fit(X_train, y_train)
```

```
▾ DecisionTreeClassifier
DecisionTreeClassifier()
```

PREDICTION:

```
dt_pred = dt_model.predict(X_test)
print(dt_pred[:10])
```

```
[0. 0. 0. 0. 0. 0. 0. 0. 0. 1.]
```

ACCURACY:

```
from sklearn.metrics import accuracy_score
dt_accuracy = accuracy_score(y_test, dt_pred)
print("Decision Tree Accuracy:", dt_accuracy)
```

```
Decision Tree Accuracy: 0.7981512141280354
```

Accuracy=79%

The model performed reasonably well but showed lower accuracy due to overfitting.

CONFUSION MATRIX:

```
from sklearn.metrics import confusion_matrix
dt_cm = confusion_matrix(y_test, dt_pred)
print(dt_cm)
```

```
[[38172  5567]
 [ 4674  2323]]
```

38172((true negative)-correctly predicted nondiabetic cases.

5567(false positive)-predicted diabetic but actually non-diabetic.

4674(false negative)-predicted nondiabetic but diabetic.

2323(true positive)-correctly predicted diabetic cases.

The Decision Tree classifier correctly identified 2,323 diabetic patients and reduced the number of false negatives compared to Logistic Regression. Although the overall accuracy of the model was lower, it demonstrated better diabetic detection capability. From a healthcare perspective, this is important because reducing false negatives helps ensure that more diabetic individuals receive timely diagnosis and treatment. Therefore, the Decision Tree highlights the trade-off between overall accuracy and diabetic recall in imbalanced healthcare datasets.

CLASSIFICATION REPORT:

```
from sklearn.metrics import classification_report
print(classification_report(y_test, dt_pred))
```

```
              precision    recall  f1-score   support

         0.0       0.89      0.87      0.88     43739
         1.0       0.29      0.33      0.31      6997

    accuracy                           0.80     50736
   macro avg       0.59      0.60      0.60     50736
weighted avg       0.81      0.80      0.80     50736
```

The classification report provides important evaluation of metrics such as Precision, Recall, F1score, and Support. These metrics help assess the effectiveness of the Decision Tree model in predicting diabetic and non-diabetic cases.

NON DIABETIC CLASS(0):

Precision = 0.89 means 89% of cases predicted as non-diabetic were actually non-diabetic.

Recall = 0.87 means the model correctly identified 87% of all non-diabetic individuals.

F1-score = 0.8 Strong performance for the majority class.

DIABETIC CLASS(1):

Precision = 0.29 means Only 29% of predicted diabetic cases were actually diabetic.

Recall = 0.3 means the model identified 33% of actual diabetic patients.

F1-score = 0.31  Moderate performance for diabetic detection.

Macro Average:

The macro average calculates the average performance across all classes by giving equal importance to both diabetic and non-diabetic classes, regardless of the number of samples.

For the Decision Tree model:

Precision = 0.59

Recall = 0.60

F1-score = 0.60

These values indicate that the model provides a moderate and relatively balanced performance across both classes. The higher recall compared to Logistic Regression suggests that the Decision Tree is more effective in identifying diabetic patients.

Weighted Average:

The weighted average calculates the performance metrics by considering the number of samples present in each class.

For the Decision Tree model:

Precision = 0.81

Recall = 0.80

F1-score = 0.80

Since the dataset contains a larger number of non-diabetic records, the weighted average is influenced more by the performance of the non-diabetic class. Therefore, the weighted average values are higher than the macro average values.

## RANDOM FOREST:

Random Forest is an machine learning algorithm that combines multiple decision trees.

Random Forest was selected because:

Reduces overfitting.

Improves prediction stability.

MODEL TRAINING:

```
rf_model.fit(X_train, y_train)
```

```
RandomForestClassifier
RandomForestClassifier(random_state=42)
```

PREDICTION:

```
rf_pred = rf_model.predict(X_test)
print(rf_pred[:10])
```

```
[0. 0. 0. 0. 0. 1. 0. 0. 0. 1.]
```

ACCURACY:

```
from sklearn.metrics import accuracy_score
rf_accuracy = accuracy_score(y_test, rf_pred)
print("Random Forest Accuracy:", rf_accuracy)
```

```
Random Forest Accuracy: 0.8598628192999054
```

Accuracy~86%

Random forest provided reliable and stable predictions.

CONFUSION MATRIX:

```
from sklearn.metrics import confusion_matrix
rf_cm = confusion_matrix(y_test, rf_pred)
print(rf_cm)
```

```
[[42416  1323]
 [ 5787  1210]]
```

42416(true negative)-correctly predicted nondiabetic cases.

1323(false positive)-predicted diabetic but actually non-diabetic.

5787(false negative)-predicted nondiabetic but diabetic.

1210(true positive)-correctly predicted diabetic cases.

The Random Forest confusion matrix shows that the model correctly classified 42,416 non-diabetic individuals and 1,210 diabetic individuals. However, it incorrectly classified 5,787 diabetic patients as non-diabetic, resulting in a large number of false negatives. Although the model achieved high overall accuracy, its ability to detect diabetic cases was limited. From a healthcare perspective, this is a significant concern because missed diabetic patients may remain undiagnosed and untreated. Therefore, confusion matrix analysis reveals that high accuracy does not necessarily indicate effective diabetic detection(even if the model's accuracy is high, it may still be poor at finding diabetic patients).

CLASSIFICATION REPORT:

```
from sklearn.metrics import classification_report
print(classification_report(y_test, rf_pred))
```

```
              precision    recall  f1-score   support

         0.0       0.88      0.97      0.92     43739
         1.0       0.48      0.17      0.25      6997

    accuracy                           0.86     50736
   macro avg       0.68      0.57      0.59     50736
weighted avg       0.82      0.86      0.83     50736
```

The classification report was used to evaluate the performance of the Random Forest classifier. It provides important metrics such as Precision, Recall, F1-score, and Support for each class.

NON DIABETIC CLASS (0):

Precision = 0.88 means 88% of cases predicted as non-diabetic were actually non-diabetic.

Recall = 0.97 means the model correctly identified 97% of all non-diabetic individuals.

F1-score = 0.92  excellent performance for the non-diabetic class.

DIABETIC CLASS (1):

Precision = 0.48 means 48% of predicted diabetic cases were actually diabetic.

Recall = 0.17 means the model detected only 17% of actual diabetic patients.

F1-score = 0.25 Poor performance in detecting diabetic individuals.

MACRO AVERAGE

The macro average gives equal importance to both diabetic and non-diabetic classes regardless of the number of samples.

Precision = 0.68

Recall = 0.57

F1-score = 0.59

The lower macro average values indicate that the model's performance is not balanced across both classes, particularly due to poor diabetic detection.

WEIGHTED AVERAGE:

The weighted average considers the number of samples in each class while calculating the overall metrics.

Precision = 0.82

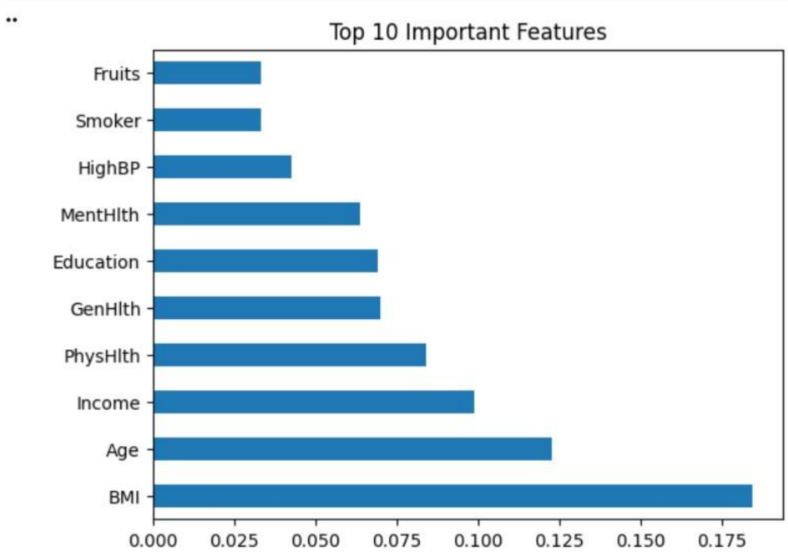
Recall = 0.86

F1-score = 0.83

The higher weighted average values are influenced by the large number of non-diabetic records in the dataset.

## **11.FEATURE IMPORTANCE ANALYSIS:**

```
feature_importance = pd.Series(rf_model.feature_importances_,index=X.columns)
feature_importance.nlargest(10).plot(kind='barh')
plt.title("Top 10 Important Features")
plt.show()
```




BMI, Age, HighBP, and GenHlth were identified as the most influential features affecting diabetes prediction.

The Random Forest feature importance analysis identified BMI, Age, HighBP, and GenHlth as the most influential variables for diabetes prediction. Feature importance indicates how much each feature contributes to the model's decision-making process.

BMI (Body Mass Index):

BMI emerged as one of the most important features in the model. This finding is medically meaningful because obesity is a major risk factor for Type 2 Diabetes. Excess body fat increases insulin resistance, reducing the body's ability to regulate blood glucose levels effectively. Therefore, individuals with higher BMI values have a greater likelihood of developing diabetes.

Age:

Age was identified as another important feature. Diabetes prevalence generally increases with age due to reduced insulin sensitivity, metabolic changes, and long-term exposure to unhealthy lifestyle factors. Older individuals are therefore more likely to develop diabetes than younger individuals.

HighBP (High Blood Pressure):

High blood pressure was found to be a significant predictor of diabetes. Hypertension frequently occurs alongside diabetes because both conditions are associated with obesity, poor diet, physical inactivity, and metabolic disorders. The model's identification of HighBP as an important feature is consistent with clinical observations.

GenHlth (General Health):

General health status was also highly influential. Individuals reporting poor general health are more likely to have chronic illnesses, unhealthy lifestyle habits, and reduced physical well-being. These factors increase the risk of diabetes and explain the importance of GenHlth in the prediction model.

Medical Knowledge:

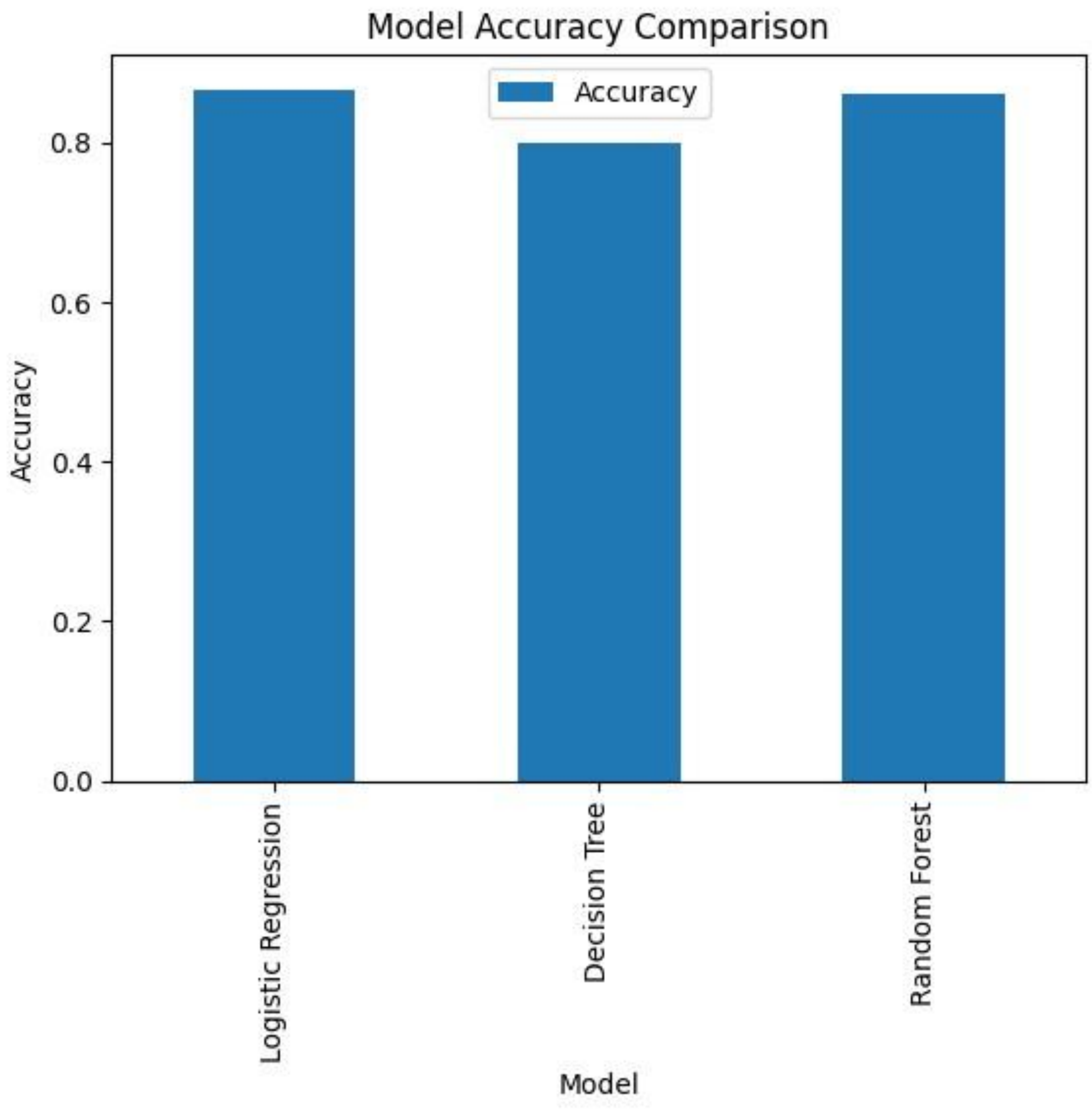
The feature importance results strongly agree with established medical understanding of diabetes risk factors. Obesity, aging, hypertension, and poor overall health are widely recognized contributors to diabetes development. The fact that the Random Forest model identified these variables as the most important features suggests that the model is learning meaningful and clinically relevant patterns from the dataset.

Overall Interpretation:

The feature importance analysis demonstrates that the machine learning model is not relying on random variables. Instead, it focuses on medically significant health indicators that are known to influence diabetes risk. This increases confidence in the reliability and practical relevance of the prediction model.

**12.COMPARISION GRAPH:**

```
comparison.plot(x='Model', y='Accuracy', kind='bar')
plt.title("Model Accuracy Comparison")
plt.ylabel("Accuracy")
plt.show()
```




MODEL COMPARISION :

Logistic Regression 86%

Decision Tree 79%

Random Forest 86%

- Logistic Regression achieved high accuracy and performed well for binary classification.
- Decision Tree produced lower accuracy due to overfitting.
- Random Forest achieved similar accuracy to Logistic Regression while providing better feature analysis capabilities.

Comparative Performance Analysis

The low recall suggests that a significant proportion of diabetic individuals were incorrectly classified as non-diabetic, resulting in a large number of false negatives. In healthcare prediction systems, false negatives are particularly critical because undetected diabetic patients may not receive timely diagnosis, monitoring, or treatment, thereby increasing the risk of disease progression and associated complications.

In contrast, the Decision Tree classifier achieved a lower overall accuracy of approximately 80%, but demonstrated improved diabetic case detection with a recall value of approximately 0.33. This indicates that the Decision Tree model was able to correctly identify a greater proportion of diabetic individuals compared to Logistic Regression and Random Forest.

These results highlight the trade-off between overall classification accuracy and minority-class detection performance in imbalanced healthcare datasets. While Logistic Regression and Random Forest achieved superior accuracy by correctly classifying a large number of non-diabetic instances, their ability to identify diabetic cases remained limited. Therefore, model evaluation should not depend only on accuracy. Additional performance metrics such as Precision, Recall, F1-score, Macro Average, Weighted Average, and Confusion Matrix analysis must be considered to obtain a comprehensive assessment of model effectiveness, particularly in medical diagnostic applications.

From a healthcare perspective, maximizing recall for the diabetic class is often more important than achieving the highest overall accuracy, as early identification of diabetic patients is essential for effective clinical intervention and disease management.

## 13.CONCLUSION:

This project successfully implemented three supervised machine learning algorithms for diabetes prediction using the CDC Diabetes Health Indicators dataset.

Exploratory Data Analysis was performed to understand feature relationships and dataset characteristics. Logistic Regression, Decision Tree, and Random Forest models were trained and evaluated using multiple performance metrics.

Among the implemented models, Logistic Regression and Random Forest achieved the highest performance. The study demonstrates that machine learning techniques can effectively support healthcare prediction systems and assist in early diabetes risk assessment.

This project successfully applied Logistic Regression, Decision Tree, and Random Forest algorithms for diabetes prediction using the CDC Diabetes Health Indicators dataset. Exploratory Data Analysis, correlation analysis, feature importance analysis, confusion matrices, and classification reports were used to evaluate model performance and identify important health indicators associated with diabetes.

The results showed that Logistic Regression and Random Forest achieved higher overall accuracies of approximately 86%, while the Decision Tree achieved an accuracy of approximately 80%. However, the evaluation revealed that accuracy alone is not a sufficient performance metric due to the significant class imbalance present in the dataset.

Although Logistic Regression and Random Forest produced high accuracy, both models exhibited low recall values for the diabetic class (approximately 0.17), indicating that many diabetic individuals were incorrectly classified as non-diabetic. This resulted in a large number of false negatives, which is a serious limitation in healthcare applications because undetected diabetic patients may not receive timely diagnosis and treatment. On the other hand, the Decision Tree model achieved lower accuracy but demonstrated better diabetic detection capability with a recall of approximately 0.33.

Despite achieving satisfactory overall performance, the current models have limitations :

- The dataset is imbalanced, with non-diabetic cases significantly more non diabetic records than diabetic records.

- Logistic Regression and Random Forest exhibited low recall for the diabetic class, resulting in a high number of false negatives.

- Accuracy alone does not provide a complete assessment of model performance in healthcare applications.

- The models were trained using survey-based health indicators and do not include clinical measurements such as blood glucose or levels.

- Model performance may vary when applied to different populations or real-world healthcare environments.

## 14.REFERENCE:


1. UCI Machine Learning Repository

2. CDC BRFSS Dataset Documentation